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Optimizing cementitious mortars for lunar construction: a multi-criteria decision-making approach

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This study investigates eight cementitious materials for lunar construction to support sustainable habitats in the Moon's extreme environment. The materials, geopolymer, sintered material, polymer-bound regolith, regolith-based magnesium oxychloride, Ordinary Portland cement (OPC), sulfur concrete, microbial-induced calcite precipitation, and geo-thermite products, were evaluated using fifteen criteria, including availability, water use, curing time, temperature resilience, strength, durability, and radiation shielding. The Entropy Weight Method (EWM) was used to assign objective weights, identifying temperature as the most critical factor. Multi-criteria decision-making (MCDM) techniques, TOPSIS, VIKOR, and WASPAS, ranked materials based on normalized data and stakeholder input. Geopolymer and sintered material emerged as top performers, followed by polymer-bound regolith and magnesium oxychloride. OPC, sulfur concrete, microbial calcite, and geo-thermite ranked lower. This rigorous, transparent framework enables informed selection of efficient, durable materials for lunar infrastructure, advancing long-term space exploration. The methodology is adaptable and minimizes bias, offering a reliable path toward resilient extraterrestrial construction.

Keywords: Lunar regolith; geopolymer concrete; sintering method; lunar habitat construction

1. Introduction

The idea of space colonization has fascinated humanity for decades. Nowadays, about 50 years after humans first ventured into space, we are seriously considering both short-term and long-term stays beyond Earth. Scientists and investors view the Moon and Mars as promising candidates for establishing (semi-)permanent human settlements. However, building habitats on these celestial bodies will require leveraging natural resources, as transporting large amounts of construction materials from Earth is neither practical nor economical. This makes *in situ* resource utilization (ISRU) a critical component of long-term planetary missions. On the Moon, locally available resources will be essential for constructing shelters, launch and landing pads, transportation routes, and factories. Yet, the Moon's environment is vastly different from Earth's, requiring researchers to develop innovative methods and materials or adapt traditional construction techniques to meet these unique challenges.

The Moon's gravity is about 16.6% of Earth's, and its atmospheric pressure is an extremely low 3.03 Pa. Its surface is highly vulnerable to meteorite impacts and exposure to harmful radiation [1,2]. Moreover, temperatures on the Moon vary dramatically depending on location, season, and time of day, ranging from a freezing −173 °C to a scorching 127 °C. These extreme and unpredictable conditions demand the use of specialized construction materials and techniques to ensure the safety and survival of humans on the lunar surface [3].

The difference in time cycles between Earth and the Moon is a crucial factor in designing space habitats. Earth completes one rotation on its axis in 24 h and orbits the Sun in about 365 days. In comparison, the Moon's rotation takes approximately 655.7 h, and it completes one orbit around the Sun in 354.4 Earth days—a lunar year [3]. Unlike Earth, where a day consists of 24 h evenly split between daylight and darkness, the Moon experiences much longer periods of light and dark. A lunar day lasts about 14 Earth days, with one side of the Moon facing the Sun, followed by a lunar night of equal length when the Sun illuminates the opposite side. Additionally, the Moon takes roughly 29 days to complete an orbit around Earth, traveling along the same general path around the Sun as Earth. These unique cycles must be carefully considered when planning for sustainable human presence on the Moon [4].

To ensure the safety and success of constructing habitats in space, engineers must tackle the challenges posed by extreme conditions. These challenges demand rethinking materials, additives, mixing processes, casting, compacting, and curing methods to minimize both foreseeable and unforeseen risks during and after construction [5]. The materials used must meet stringent requirements, such as functioning effectively in a low-temperature range to withstand the Moon's extreme cold, exhibiting

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Table 1. Earth and moon main condition.

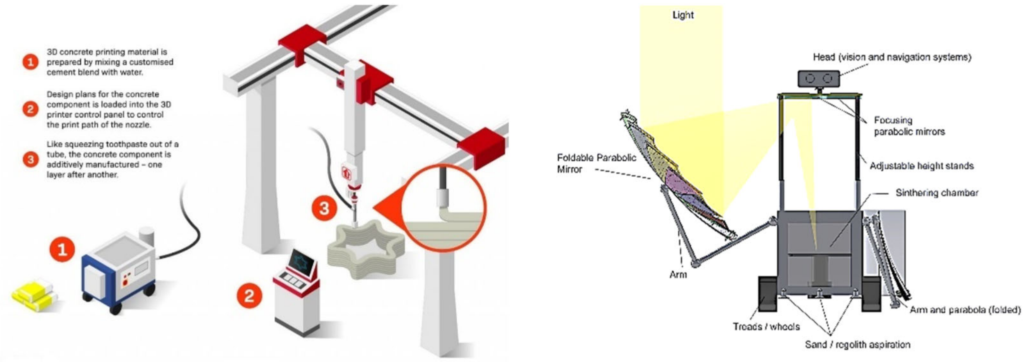
Feature	Earth	Moon	Similarities	Differences
Size	Diameter: ~12,742 km	Diameter: ~3,474 km	Both are spherical in shape due to gravity	Earth is about 4 times larger than the Moon
Gravity	9.8 m/s^2	1.62 m/s^2	Both have gravitational fields	Moon's gravity is ~1/6th of Earth's gravity
Atmosphere	Dense atmosphere with 78% nitrogen, 21% oxygen	Almost no atmosphere (very thin exosphere)	Both have extremely thin outer layers	Earth has a breathable atmosphere, Moon does not
Surface Temperature	Average: ~15 °C (varies widely)	−173 °C to 127 °C (extreme variation)	Both experience temperature changes	Moon has far more extreme temperature swings
Rotation Period	24 h (1 day)	27.3 Earth days (synchronous with orbit)	Both rotate on their axes	Moon's rotation period is much longer
Orbit	Orbits the Sun in 365.25 days	Orbits Earth in ~27.3 days	Both follow elliptical orbits	Earth orbits the Sun, Moon orbits Earth
Water Presence	70% surface covered with liquid water	Water ice in craters, no liquid water	Both have water (Earth: liquid, Moon: ice)	Earth has abundant liquid water, Moon has ice
Magnetic Field	Strong magnetic field	Weak or almost no magnetic field	Both have some form of magnetism	Earth's magnetic field is much stronger
Surface Features	Varied with mountains, oceans, plains	Craters, plains (maria), and highlands	Both have impact craters and plains	Earth has diverse terrain; Moon lacks oceans
Age	~4.5 billion years	~4.5 billion years	Both formed around the same time	None (formed from a common event)
Day/Night Cycle	24 h	~29.5 Earth days	Both experience day and night	Day on Moon lasts 14 Earth days
Life	Supports a variety of life forms	No known life	Both are in the habitable zone of the Sun	Earth supports life, Moon does not

corrosion resistance to endure the presence of reactive gases in the lunar atmosphere, and having non-ferromagnetic properties to avoid interference from magnetic dust. Additionally, these materials must provide radiation shielding [6]. Construction materials are typically tested under laboratory simulation conditions to identify potential weaknesses. However, a major hurdle is the limited availability of genuine lunar soil for testing. To address this, lunar regolith simulants are commonly used in experimental studies [7]. These simulants are designed to mimic the key characteristics of lunar soil, including mineralogical composition, morphology, and particle size distribution. Table 1 outlines the similarities and differences between Earth's and the Moon's primary environmental conditions [8–13].

The construction method is a key factor in selecting the most suitable mortar for space construction. Two main approaches proposed are 3D printing and the use of geopolymers or sintered bricks, as illustrated in Figure 1. Isachenkov, Chugunov, Akhatov, and Shishkovsky conducted a review comparing recent advancements in the chemical and technological aspects of 3D printing with lunar regolith simulants [16]. Their work outlined essential requirements and techniques for adapting additive manufacturing (AM) to space conditions [17,18]. While simple bricks can be easily produced by casting materials into molds, AM methods offer the flexibility to create more complex geometries. Beyond 3D printing and brick construction, other potential avenues for space construction remain largely unexplored, such as building underground habitats or utilizing underwater (ice) structures. These alternative methods may present innovative solutions for future space exploration challenges [8,17,19].

1.1. Binders for space habitat construction

In the construction industry, various types of concrete are produced using diverse materials and methods. However, to maintain their mechanical properties in the Moon's harsh environment, these concretes must meet both primary and secondary conditions. Primary conditions include the construction method, as well as the required temperature and pressure settings during construction and the early stages of curing. Secondary conditions involve maintaining consistent temperature and pressure until curing is complete. Binders commonly discussed for space construction include Ordinary Portland Cement (OPC), alkali-activated cement (AAC), geopolymer cement (GC), elemental sulfur (ES), magnesium- and silicon-based binders (MSBB), and water (utilized by freezing). Additionally, novel materials have been proposed, such as sintered materials, polymer-bound regolith, geo-thermite reaction products, microbial-induced calcite precipitation, and regolith-based magnesium oxychloride. However, the production of these binders requires careful consideration of energy demands and the availability of raw materials [20–22].



A B

Figure 1. 3D printing (a) and solar sintering brick (b) [14,15].

1.2. MCDM (Multiple-criteria decision-making)

Multi-Criteria Decision Making (MCDM) is an area of operations research that focuses on making decisions when multiple, often conflicting criteria are involved. Decision-making becomes particularly complex in fields like medicine, government, and business, due to the inconsistency of these criteria, some of which are quantitative and others qualitative. In the context of lunar habitat construction, engineers must account for various inconsistent criteria to select the most suitable materials and concretes. MCDM methods can help by considering both technical and economic factors. These criteria can range from qualitative aspects, such as limitations, shipping, sustainability, and safety, to quantitative measures, like temperature, curing time, strength, and density. By addressing multiple criteria, optimization approaches offer more reliable and accurate decisions than relying on a single factor. There are various MCDM methods, each with its own complexities and underlying assumptions. These differences stem from the selection of scoring scales and weights, which can create uncertainty about the reliability of the results. The complexity of MCDM methods is influenced by their statistical characteristics and mathematical processes. Key assumptions in these methods include the weighting of criteria, their relative importance, and the rationale for comparing rows and columns in a matrix [23,24].

MCDM methods can be classified into three groups [25–27]:

1. Value evaluation approaches: In this approach, a numerical rate is assigned to each alternative, and a weight is determined for each criterion to demonstrate its influence (e.g., Entropy).
2. Reference level and goal models: These approaches estimate how well alternatives can help achieve specific targets (e.g., TOPSIS).
3. Dominating methods: These models compare alternatives pairwise for each criterion and determine their superiority (e.g., PROMETHEE, ELECTRE).

MCDM methods have been successfully used across various fields, including military, petroleum and civil engineering, medicine, and space exploration [28,29]. In the field of aerospace engineering, Tavana [24] applied MCDM to assess the risks and benefits of three different mission architecture options for Mars exploration.

Gunaydin, Duvan, and Ozceylan [30] applied the MCDM method to identify the best habitable planet among nine potential residential planets, including the Moon. Their criteria encompassed a range of factors, such as gravity, mass, escape velocity, diameter, density, rotation time, distance from the Sun, length of the day, perihelion, aphelion, orbital period, orbital inclination, orbital velocity, orbital elongation, mean temperature, obliquity to orbit, and the number of satellites. The Analytic Hierarchy Process (AHP) was then used to determine the optimal properties for a lunar construction habitat [31]. Higgins and Benaroya also employed AHP to define the best approach for lunar habitat construction. Additionally, Sánchez-Lozano, Moya, and Rodríguez-Mozos [32] utilized MCDM methods to rank and prioritize exoplanets in the search for biomarkers. Even Eren and Katanalp [33] applied the MCDM method to evaluate potential sites for Bike-Sharing Systems (BSSs), considering transportation and recreational land uses.

To the best of the authors' knowledge, there has been no technical review of the proposed space binder materials, nor a comprehensive comparison of these materials. Additionally, the impact of lunar conditions on the selection of suitable cement mortars has not been explored. The main objectives of this research are to compare and summarize space mortar concretes, evaluate the weight of various criteria (such as temperature, shipping, etc.), and determine the best material considering inconsistent criteria. To achieve this, the study reviews eight key construction materials based on 15 important criteria. The Entropy method is used to assign weights and identify the most significant criteria, while Multiple-Criteria Decision-Making methods are applied to determine the best mortars for lunar habitat construction. Three MCDM methods—Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR), and Weighted Aggregates Sum Product Assessment (WASPAS)—are employed in this study. This study also explores the differences in the properties of lunar regolith, including its mineralogy and morphology, to identify the most suitable

Figure 2. The process of utilizing MCDM techniques to choose the optimal concrete mortar for planetary applications.

concrete material for the Moon. In particular, We compare the properties of materials like; geopolymer, sulfur concrete, and sintered materials untill determine which is best suited for lunar construction.

2. Methodology

To determine the ideal concrete mortar for space habitat construction, the primary candidate materials were first discussed, highlighting their advantages and disadvantages. Key criteria related to concrete mortars were then explained. Finally, the Entropy method was used alongside three MCDM techniques like TOPSIS, VIKOR, and WASPAS. Figure 2, illustrates the process of using these MCDM methods to select the best space concrete mortar.

2.1. Construction materials

This review examines the primary materials proposed for constructing space habitats, including ordinary Portland cement, sulfur concrete, geopolymer, sintered material, polymer-bound regolith, geo-thermite reaction products, microbial-induced calcite precipitation, and regolith-based magnesium oxychloride. The key characteristics, advantages, and disadvantages of each material are also discussed. The eight construction methods for lunar habitats have distinct advantages and disadvantages. Ordinary Portland Cement (OPC) is a proven, strong, and durable Earth-based technology but requires significant water, scarce on the Moon, and is hard to produce in lunar conditions. Sulfur concrete uses abundant lunar sulfur, needs no water, and forms a binder when melted, but it's brittle, temperature-sensitive, and has low tensile strength. Geopolymer concrete offers high strength and durability using lunar regolith with less energy than OPC, yet requires hard-to-source alkali activators. Sintered materials fuse lunar regolith with solar or microwave energy without binders but are energy-intensive and yield lower strength. Polymer-bound regolith concrete creates flexible, lightweight structures with lunar regolith but relies on imported polymers and may degrade over time. Geo-thermite reaction products use lunar elements for strong, heat-resistant materials, though the process is complex and energy-heavy. Regolith-based magnesium oxychloride is fast-setting and strong, using abundant lunar magnesium, but needs scarce chlorides and is moisture-sensitive during curing. Microbial-induced calcite precipitation (MICP) is eco-friendly and self-healing with lunar materials but requires a controlled biological environment, nearly impossible on the Moon. Geopolymer concrete and sintered materials are promising for leveraging lunar resources, but each method balances resource use, energy needs, and lunar environmental challenges differently. Table 2 provides a comparison of the benefits and drawbacks of these methods [9,34–42].

2.1.1. Ordinary portland cement (OPC)

Ordinary Portland cement (OPC) is a hydraulic cement produced by mixing limestone ($CaCO_3$) with Si-rich clay or shale and heating the mixture to around 1450 °C [10,43]. Known for its high durability, strength, and simplicity, OPC has been a primary focus of space construction research as a terrestrial binder [44–46]. However, using OPC in space presents several challenges. First, the Moon lacks significant quantities of $CaCO_3$, which is essential for OPC production. Second, water is required for the cement's production, hydration, and curing processes. While liquid water is not stable on the Moon's surface, it may be found beneath the surface or produced by melting ice. Additionally, the curing time for OPC on Earth is typically 28 days under specific temperature and pressure conditions. Special measures would be needed to prevent the freezing or evaporation of water during hydration in the Moon's low temperatures and vacuum pressure [47]. On the positive side, OPC could offer effective shielding against galactic cosmic rays (GCR) and solar particle events (SPE) [48,49].

2.1.2. Sulfur concrete

Sulfur binder has been considered a promising option for space concrete [45,50]. It can be produced from sulfate minerals, such as ferric sulfate ($Fe_2(SO_4)_3$), which are abundant on the lunar surface [51,52]. The production process involves two key steps: first, synthesizing sulfur through the pyrolysis of ferric sulfate at approximately 900 °C, and second, catalytically reducing the sulfur dioxide (SO_2) gas produced during the process [53–58]. To synthesize one kilogram of ore, 22.5 grams of hydrogen and 105 grams of CO are needed, both of which can be produced on the Moon from water and CO_2, respectively. The reaction yields valuable byproducts, including 240 grams of elemental sulfur, 203 grams of water, and 120 grams of oxygen. Waste products like 400 grams of Fe_2O_3, 165 grams of CO_2, and additional water and carbon dioxide can be reused to regenerate the hydrogen and CO. For use as cement, elemental sulfur is mixed with grains while molten (above 120 °C) for casting and 3D printing. The mortar sets and hardens rapidly when the temperature drops to 112.8 °C. This sulfur-based binder is a promising option for space construction as it utilizes local resources and does not require water for mixing or

Table 2. Advantage and disadvantage of methods.

Construction Method	Advantages	Disadvantages
Ordinary Portland Cement (OPC)	• Familiar, proven technology on Earth • Strong and durable structures • Readily available on Earth	• Requires significant water, which is scarce on the Moon • Difficult to produce cement in lunar conditions
Sulfur Concrete	• Abundant on the Moon • Can be used as a binder when melted and cooled • Requires no water	• Brittle and sensitive to temperature changes • Low tensile strength, may need reinforcement
Geopolymer Concrete	• High strength and durability • Can be made using lunar regolith • Less energy-intensive than traditional cement	• Requires alkali activators (e.g. sodium or potassium), which may be hard to source on the Moon
Sintered Materials	• Involves using solar or microwave energy to fuse lunar regolith • No need for additional binding agents	• Energy-intensive process • Results in limited structural strength compared to other materials
Polymer-bound Regolith Concrete	• Can use lunar regolith as filler • Flexible and can form lightweight, durable structures	• Requires importing polymers, which increases payload weight • May degrade over time under lunar conditions
Products of Geo-Thermite Reactions	• Uses lunar elements for reactions • Potentially strong, metallic end products • High heat resistance	• Complex chemical process • Requires precise control and substantial energy input
Regolith-based Magnesium Oxychloride	• Can be made with magnesium, which is abundant in lunar regolith • Fast-setting and strong	• Requires chloride source, which might be scarce on the Moon • Sensitive to moisture during curing
Microbial Induced Calcite Precipitation (MICP)	• Environmentally friendly • Can create self-healing structures • Uses local lunar materials	• Requires a controlled biological environment • Extremely challenging to maintain on the Moon due to extreme conditions

casting. Furthermore, it can be easily recycled by heating and recasting. Despite its favorable mechanical and chemical properties for space applications, the main challenge lies in the synthesis of sulfur itself [11,59].

2.1.3. *Geopolymer concrete*

Geopolymers are typically made from by-products like fly ash and slag, offering an eco-friendly alternative to traditional cement. They are known for their high uniaxial and flexural strength, excellent durability, and low carbon emissions [60–63]. These binders are composed of silico-aluminates arranged in an amorphous to semi-crystalline 3D structure. Geopolymers offer several benefits, including fast and controllable setting and hardening times, high compressive strength, resistance to freeze-thaw cycles, excellent durability in sulfate and acidic environments, fire resistance with low thermal conductivity, low shrinkage, and effective radiation shielding [64–69]. A key drawback of geopolymer concrete is its need for water. However, a study by Wang, Tang, Cui, He, and Liu [70] proposed a method to recover water after curing using tektite powder. With a low operating temperature (70–90 °C) and a curing time of just one day, geopolymers are well-suited for space construction. In comparison, Portland cement requires 28 Earth days to cure, while sulfur cement and sintered materials require higher temperatures for curing [71,72].

2.1.4. *Sintered materials*

Construction materials produced through thermal processes can be broadly categorized into two types: sintered materials and those made through melting. Common sintering methods include laser, radiant furnace, and microwave sintering [73,74]. Sintering is a thermal treatment technique that fuses particles into a solid without fully melting them, achieved through atomic-scale mass transport [75]. This process reduces the overall surface area of grains by bonding them together, creating new interfaces between the particles [76]. When lunar regolith is heated near its melting point, it transforms into a sintered metal oxide material, where the particle surfaces diffuse into each other [17,77,78]. Sintering requires less energy than melting and does not need additives or additional processes [79]. Examples of sintered materials include ancient ceramic pots and modern composites. Solar sintering, a cutting-edge space technology, has been proposed for lunar concrete production [80,81]. In contrast, melting involves liquefying regolith above its melting temperature in casting processes. This method can produce dense materials with high strength and wear resistance. Depending on the degree of crystallization, these materials can be classified as glass, glass-ceramic, or cast [82]. The main disadvantage of melting and casting is their high energy consumption due to the required temperatures, while the advantage is that no additives are needed [83].

2.1.5. Polymer-bound regolith concrete

Hintze, Curran, and Back [76] were pioneers in using polymers to bind lunar regolith simulants, creating a polymer/regolith composite with less than five mass percent polymer. Polymers needed for in-situ production can be extracted from volatiles found on planetary bodies or obtained from asteroids or moons containing kerogen-like organic material [12,76,84]. Researchers like Chen [85] have explored polymer concrete, developing a polymer with a high strength-to-weight ratio and strong radiation resistance, which can effectively bind lunar regolith grains together. Johnson-Freese [86] highlighted the potential of polymers as a promising material for use in space, citing their ability to act as radiation shields, as well as their role in chain scission and crosslinking. However, a key drawback of this concrete is its relatively low strength.

2.1.6. Products of geo-thermite reactions

The geo-thermite reaction is an oxidation-reduction process that involves a mixture of minerals, glass, and aluminum. It is a self-propagating, high-temperature synthesis (SHS) reaction, where chemical reactions between intact aggregates and a reducing agent exhibit thermite-like behavior. This reaction shows significant potential for in-situ resource utilization (ISRU) in space construction. The heat generated during the geo-thermite reaction can melt at least one of the reactants, allowing them to bind together [87–89]. However, further research is needed to explore its viability as a cement for space habitats. Two major limitations of this method are the energy-intensive extraction process and the large amount of metal required for the powder, which pose challenges for space construction [22].

2.1.7. Regolith-based magnesium oxychloride

Magnesium oxide (MgO) has been used on Earth for over a century in the form of Sorrel cement (magnesium oxychloride) [90]. Magnesium oxychlorides (MOCs), introduced in 1867 by Stanislas Sorel, are non-hydraulic binders that can be chemically defined as various components of the $MgO\text{-}MgCl_2\text{-}H_2O$ synthesis [91]. Studies by Jiangxiong, Yimin, and Yongxin [92], as well as Jin and Al-Tabbaa [93], have shown that MgO can be combined with amorphous silica to form magnesium-silica-hydrate (MSH), which is similar to the calcium silicate hydrate (CSH) found in traditional Portland cement. Tran and Scott [94] explored the potential of this cement for high-strength structural applications by developing a MgO-silica fume mortar with a compressive strength of 87 MPa [95]. MOC cement has a fast setting time, high initial strength, and good rheological properties, and it does not require water for curing [95]. Magnesium is abundant on the lunar surface, and there are no limitations on its in-situ utilization. Additionally, magnesium chloride ($MgCl_2$) or monopotassium phosphate (KH_2PO_4) is necessary to complete the cementation process. On the Moon, chloride ions are often found in apatite minerals, and potassium and phosphorus can be sourced from KREEP (potassium, rare Earth elements, and phosphorus) materials. Phosphate minerals such as apatite, whitlockite/merrillite, and keiviite are present on the lunar surface [45,96–98].

2.1.8. Microbially induced calcite precipitation

Biocementation, or microbially induced carbonate precipitation (MICP), uses microorganisms to precipitate calcium carbonate ($CaCO_3$) as a binding agent [99–102]. This method is commonly used for repairing damaged concrete, stabilizing soil, and as an alternative to conventional Portland cement [103,104]. Researchers have increasingly focused on the potential of MICP for *in situ* construction of Lunar habitats. In this process, urease-producing bacteria promote the precipitation of calcium carbonate between loose aggregates. Along with the bacteria, a calcium source and urea are required. Urea reacts with water to generate ammonia and carbonate, and when supersaturation occurs, calcium carbonate forms, binding the grains together [105]. Calcium can be derived from minerals like limestone, and urea can be sourced from human or animal urine in space. The only materials that need to be transported to space are the bacteria, growth media, necessary equipment, and water (if not available on the Moon). Since the effectiveness of bacteria is temperature-dependent, maintaining the proper temperature conditions is crucial for this method [106,107].

2.2. *Criteria of construction materials*

To identify the most suitable concrete mortar for constructing space habitats, fifteen criteria were chosen: availability, shipping feasibility, water requirements, technical working conditions, curing time, operating temperature, total energy consumption, strength, durability, cosmic radiation shielding (density and hydrogen content), required additives, sustainability, safety, and recyclability. While cost is a significant factor, it was excluded from consideration because the specific construction methods for lunar conditions have not yet been finalized, and their associated needs remain unclear [108]. Furthermore, since the composition of lunar regolith has not been comprehensively analyzed for all types of concrete, it was not included as a criterion in the MCDM process. Instead, only the differences in regolith properties for geopolymer, sulfur, and sintered materials were examined.

2.2.1. Availability

The construction of permanent planetary habitats is heavily influenced by the efficiency of building materials and the availability of in-situ resources [109]. Utilizing locally sourced materials significantly reduces transportation and equipment costs, along with other logistical expenses. Over the past decades, engineers and economists have studied the challenge of material availability [110]. For instance, while lunar soil is plentiful, resources like water are scarce. Organic materials, such as polymers, have not been identified on the Moon, but minerals like silicon oxide, iron oxide, and aluminum oxide are abundant. In contrast, carbonates are rare, which greatly impacts the choice of construction methods and mortar types. To address this, researchers are focusing on leveraging in-situ materials [111]. In conclusion, concrete made from materials readily available on the Moon's surface, without extensive extraction or processing, offers a clear advantage. This approach enhances construction efficiency, reduces costs, and provides a sustainable and practical solution for building permanent planetary habitats.

2.2.2. *Shipping*

Shipping is a critical factor in lunar construction due to the exorbitant cost, which is approximately $130,000 per kilogram. Some proposed materials, such as polymers or bacteria, require specialized resources to be transported from Earth [86,87,112]. Given these high shipping costs, the focus has shifted toward utilizing in-situ resources for civil infrastructure. Materials like sintered regolith, which eliminate the need for material transfer from Earth, are particularly appealing to researchers. Building with locally sourced materials not only reduces costs but also makes the establishment and maintenance of lunar structures more feasible, paving the way for a sustainable and permanent presence on the Moon.

2.2.3. *Water requirement*

Water is a critical factor in the selection of materials for constructing space habitats. Certain materials, like Portland cement, rely heavily on water for hydration, whereas alternatives such as sulfur concrete do not require any water [113]. In the context of space construction, materials that eliminate the need for water hold a significant advantage. Water is an invaluable and scarce resource in space, and transporting it from Earth to the Moon or other celestial bodies is both expensive and logistically challenging. Consequently, using water-free materials simplifies construction, reduces costs, and enhances the feasibility of establishing a sustainable, long-term presence on the Moon or other extraterrestrial environments.

2.2.4. *Technical working conditions*

The effective use of construction mortars depends heavily on specific technical working conditions. These include processing raw materials, storage, transportation to the construction site, and the requirements during and after construction, such as printing and curing conditions. For example, Portland cement mortar requires careful control during printing to prevent water in the mix from freezing under low temperatures or evaporating in a vacuum. Additionally, it has a 28-day curing period under controlled conditions to achieve full strength, which necessitates prolonged monitoring and preparation. The production process for this cement also involves significant effort to convert raw materials into usable cement [114]. In contrast, sulfur concrete must be maintained above 120 °C to stay molten, but it solidifies quickly after printing and does not require curing [113]. Materials with simpler printing and curing requirements, such as sintered materials, or those that do not demand specialized facilities, are generally more appealing to engineers [15]. Therefore, the technical working conditions associated with construction mortars play a crucial role in determining their practicality and feasibility for building space habitats.

2.2.5. *Curing time*

Curing time is a critical factor in choosing the right construction mortar for space habitats. After a 3D printer lays down a layer of mortar, the material must cure quickly to support subsequent layers and maintain structural integrity [115]. The curing time also needs to be optimized to ensure the mortar solidifies properly in the pipes before being extruded through the nozzle. This time is influenced by the rheological properties of the mortar and, in some cases, can be adjusted using retarders or accelerators. For instance, Portland cement mortar requires 28 Earth days to achieve its full strength, whereas materials like sulfur concrete and sintered materials solidify and reach their maximum strength almost immediately after cooling [116,117]. Mortars with shorter curing times simplify the construction process, requiring less equipment and fewer resources. As a result, curing time is a significant consideration in space construction, as it directly impacts the efficiency and practicality of the building process.

2.2.6. *Temperature*

Temperature considerations for construction mortars can be divided into two key aspects: the temperature needed for material processing and the working temperature during application [5]. For instance, producing Portland cement requires processing temperatures of around 1400 °C, while extracting sulfur from Martian regolith demands about 900 °C. During application, Portland cement mortar must be kept above the freezing point of water, whereas sulfur concrete needs to be maintained at 120 °C to stay molten for use [50]. In contrast, some materials, like sintered concrete, bypass raw material processing entirely, requiring only heating to 1400 °C to melt regolith [73]. High-temperature methods typically necessitate advanced energy sources, such as nuclear power or emerging technologies, whereas solar cells are sufficient for processes requiring lower temperatures. This study evaluates the maximum temperature needed for each material in the decision matrix. Temperature requirements are a critical factor in space construction, as they directly influence the type of equipment and energy sources needed for processing and applying the construction materials.

2.2.7. *Total required energy*

The total energy required per unit volume is a crucial consideration in constructing space habitats [118]. Some mortars demand significant energy for processing and specific conditions during curing. For example, Portland cement mortar requires processing temperatures of 1400 °C and a 28-day curing period at temperatures above the freezing point of water to achieve full hydration [46,117]. Given the limited energy resources on the Moon or other planetary surfaces, choosing materials that minimize energy consumption during processing and curing is essential. Materials with lower energy requirements can make construction more efficient and support the goal of establishing a sustainable, permanent presence on the Moon. Consequently, evaluating the total energy needed per unit volume is a key factor when selecting construction mortars for space habitats.

2.2.8. Strength

Concrete strength is one of the most vital considerations in building space habitats. The materials must not only support the structure's weight but also endure the unique stresses of the space environment. On the Moon, where gravity is weaker than on Earth, taller buildings can be constructed with relative ease. However, this reduced gravity introduces specific stresses that construction materials must withstand. Impact and flexural strength are particularly critical, as structures need to resist potential damage from meteorite impacts [55,119]. Additionally, the harsh conditions of space demand durable materials capable of maintaining integrity over time. Therefore, the strength of construction mortar directly influences the safety, stability, and longevity of space habitats. Robust materials that can endure environmental stresses while supporting structural loads are essential for establishing a permanent presence on the Moon or other planetary surfaces.

2.2.9. Durability

Durability is a crucial factor for construction mortar in the long term, as some materials may degrade under vacuum pressure or extreme temperature variations. For instance, sulfur concrete begins to sublimate at temperatures above zero degrees Celsius, which could compromise its durability in the space environment. Durability is also tied to a material's flexibility and its ability to withstand fluctuations in internal and external pressure, as well as temperature changes and meteorite impacts [120–122]. Therefore, the durability of construction mortar plays a vital role in space habitat construction. It directly impacts the safety, stability, and longevity of the structure. The materials must be resilient enough to endure the harsh conditions of space and maintain their integrity over time, ensuring the feasibility of establishing a permanent presence on the Moon or other planetary surfaces.

2.2.10. Cosmic radiation shield (density and hydrogen content)

Shielding humans and equipment from primary radiation, including galactic cosmic rays (GCR) and occasional solar particle events (SPE), is a key challenge in space exploration. This radiation poses serious risks to human health, animals, and plants, and can also damage electronic devices. Primary space radiation consists mainly of protons and other ions [123,124]. Galactic Cosmic Rays include ions with varying energy levels, while Solar Particle Events are typically composed of high-energy protons along with gamma and X-rays. In addition to primary radiation, secondary radiation is produced when high-energy particles interact with materials such as lunar habitat concrete. These interactions can create additional radiation in the form of charged particles, neutrons, and gamma rays. The level of secondary radiation depends on the type of incoming radiation and the shielding material, which can either reduce or amplify the effects of primary radiation [125]. The effectiveness of shielding is largely determined by the material's hydrogen content and its density, measured in surface density units (g/cm^2) [126,127]. Research has shown that the compaction of lunar regolith, rather than its composition, plays a larger role in radiation shielding, with the hydrogen content being a crucial factor [117,128–132]. Materials with high hydrogen content, such as water-based concretes and polymers, are particularly effective for radiation protection [86,133,134]. Thus, the ability of construction mortar to shield against both primary and secondary radiation is a vital consideration for space habitat construction. The materials used must be capable of providing sufficient protection to ensure the safety and health of the crew, making it feasible to establish a permanent presence on the Moon or other planetary surfaces.

2.2.11. Additives needed

Cement mortar additives are materials incorporated into cement to enhance its properties. These additives include accelerators, retarders, dispersants, extenders, weighting agents, and foaming agents. For example, Portland cement is formulated with a variety of additives to modify the rheological characteristics of the mortar and improve its final strength. In contrast, some mortars, such as sulfur concrete, either lack specific additives or have not been extensively studied in this regard. The inclusion of additives in construction mortar can influence key properties such as curing time, strength, durability, and radiation shielding effectiveness. Therefore, selecting and optimizing the right additives is crucial in space habitat construction, as they can significantly impact the performance and safety of the structure over time. The materials used must provide the necessary properties to withstand the challenges of the space environment, making it possible to establish a permanent presence on the Moon or other planetary surfaces.

2.2.12. Sustainability

Sustainability refers to the ability to maintain balance across various aspects of life, particularly in the context of the coexistence of Earth's biosphere and human habitats [135]. In recent decades, it has come to signify the use of renewable and natural resources to support long-term human survival [136,137]. Sustainability is typically defined by three key pillars: the environment, the economy, and society. This concept encourages the use of resources in a way that allows people to continue living without depleting them or causing harm to future generations [16,138,139].

In the context of space habitat construction, certain types of concrete may be more environmentally friendly, producing fewer pollutants during processing and printing, and generating less waste. However, from an economic perspective, some of these materials might require frequent repairs and maintenance, either during the construction phase or over time. Socially, the long-term effects of these materials have not yet been fully studied [56,140]. Therefore, sustainability plays a crucial role in space habitat construction, as it directly impacts the long-term viability of the habitat as well as its effects on the environment, economy, and society. The materials used must align with sustainability goals, ensuring that a permanent presence on the Moon or other planetary surfaces can be established without causing significant harm to these crucial areas.

2.2.13. Safety

Safety is a crucial factor in the construction of space habitats, as it directly impacts the health and well-being of the crew, as well as the overall safety of the habitat. This includes considering the effects of concrete at various stages of production and use [141–143]. Traditional materials like Portland cement, which have been in use for a long time, are generally considered safe. However, the long-term safety of newer materials remains unknown [144]. Certain materials, such as sulfur concrete, pose known risks. For example, molten sulfur above 154 °C can release harmful gases like hydrogen sulfide and sulfur dioxide, which can be dangerous to human health [56]. Geopolymer concrete, which involves using highly alkaline NaOH solutions, also requires special precautions due to potential health risks during preparation [145]. Additionally, some materials, such as sintered products and those derived from geo-thermite reactions, may emit gases during processing, necessitating safety evaluations. Therefore, safety is a key consideration in space habitat construction. The materials chosen must meet stringent safety standards to ensure the well-being of the crew. The potential hazards associated with the preparation and use of construction materials need to be carefully assessed and mitigated, ensuring that a permanent presence can be established on the Moon or other planetary surfaces without compromising the safety of the crew or habitat.

2.2.14. Recyclability

Recyclability is an important factor to consider when selecting construction materials for space habitats on the Moon, as producing each gram of these materials requires significant energy and raw materials. This factor should be carefully considered in future research. Concrete recycling typically involves crushing and removing old concrete to be repurposed for new construction [146,147]. However, recycling Portland cement is challenging due to the irreversible nature of its hydration process. In contrast, materials like sulfur concrete and sintered products are recyclable, as new concrete can be made by melting and reusing the previous cement [58,148,149]. Thus, recyclability plays a crucial role in space habitat construction because it impacts the sustainability and environmental footprint of the habitat. The materials used must be recyclable to ensure that a permanent presence on the Moon or other planetary surfaces does not deplete natural resources or harm the environment. Future research should focus on developing construction materials that are both easily recyclable and sustainable, ensuring their long-term viability for space habitat construction.

2.3. MCDM (Multiple criteria decision making)

The selection of TOPSIS, VIKOR, and WASPAS methodologies for this study was based on their relevance in both space and civil engineering, and on expert opinions and scholarly research. Several factors were taken into account when choosing each technique, including the number of alternatives and criteria, computational complexity, flexibility in decision-making, and the ability to handle uncertainty. The objective of using MCDM techniques in this study was to establish a hierarchy of options, requiring a comprehensive ranking of the alternatives. The selected method needed to consider both the benefits and drawbacks of each option, as well as quantitative and qualitative aspects [150].

The Entropy-Weight-Method (EWM) is a type of weight assignment approach that assesses the degree of dispersion of various sources of information during the decision-making process. It has been extensively applied in comprehensive evaluation research, where the weights of different indices are established based on the entropy value of each evaluation index [151].

The TOPSIS method assesses how effectively alternatives achieve specific objectives. On the other hand, the VIKOR method is employed for decision optimization and takes into account the possibility of making compromises to resolve conflicts. It ranks the alternatives and identifies the compromise solution that is closest to the ideal [152].

The WASPAS method is a practical approach that places a strong emphasis on the accuracy of rankings. It heavily relies on the concept of ranking accuracy and utilizes both the weighted sum model (WSM) and the weighted product model (WPM) to achieve this goal. By combining different elements of MCDM methods, the ranking accuracy is improved [153–155].

To utilize MCDM methods for selecting the optimal concrete mortar, a decision matrix must be created based on various criteria and alternatives. Table 3 illustrates the decision matrix and A_1, A_2, $\ldots A_n$ correspond to the possible options selected by decision-makers. C_1, C_2, $\ldots C_n$ represent the selection criteria, and X_{ij} indicates the ratio of A_i to C_j. W_j is the weight of C_j. The weight value can be calculated either *via* a direct way or from a pairwise comparison.

Table 3. Decision matrix.

	C_1	C_2	…	C_n
A_1	X_{11}	X_{12}	…	X_{11}
A_2	X_{21}	X_{22}	…	X_{2n}
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
A_m	X_{m1}	X_{m2}	…	X_{mn}
W	W_1	W_1	…	W_1

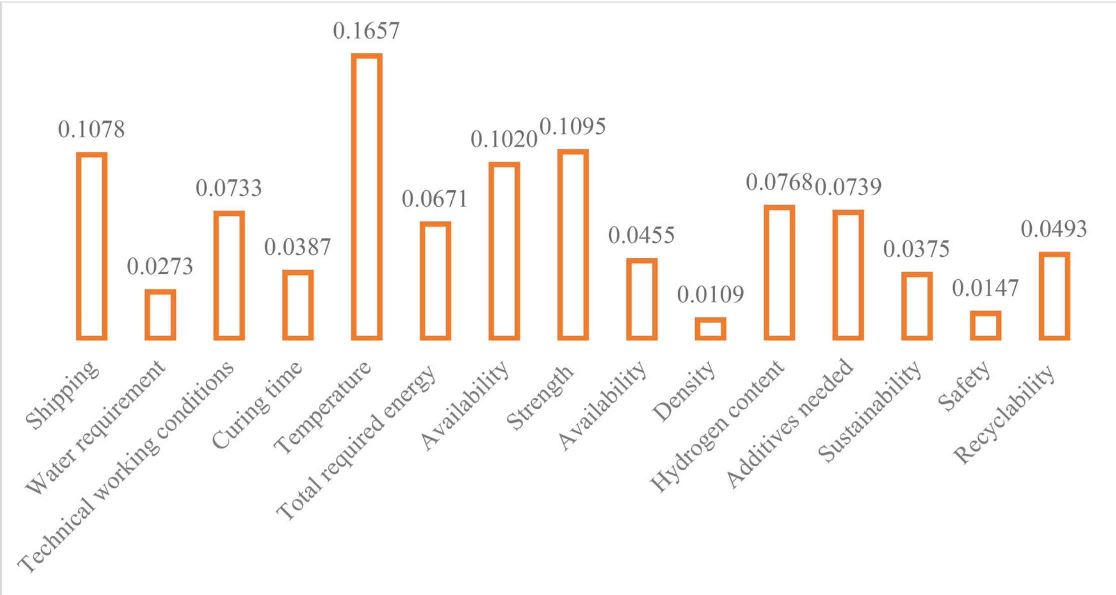


Figure 3. Weighting the parameters with the EWM method.

2.3.1. *Entropy weight method*

This method involves evaluating *n* criteria and *m* types of concrete, and the measured value of the *j*th criterias in the *i*th concretes is recorded as x_{ij}.

The initial step is to standardize measured values by calculating the standardized value of the *j*th criteria in the *i*th concrete is denoted as p_{ij}, using the following formula:

$$p_{ij} = \frac{x_{ij}}{\sum_{j=1}^{n} x_{ij}} \tag{1}$$

In the EWM, the entropy value E_i of the *j*th criteria is defined as:

$$E_j = -\frac{\sum_{j=1}^{m} p_{ij}.\ln(p_{ij})}{\ln(m)} \tag{2}$$

The entropy value E_i in the EWM ranges from 0 to 1, with a higher value indicating a greater degree of differentiation for criterion *j* and providing more information. Accordingly, higher weight is assigned to the criterion. In the EWM, the weight calculation method for w_j is determined as [156]:

$$w_j = \frac{1 - E_j}{\sum_{j=1}^{n} 1 - E_j} \tag{3}$$

As shown in Figure 3, according to EWM's results, density has the lowest weight, and temperature has the highest weight. These weights indicate the importance of each criterion. An illustrative example of the MCDM method TOPSIS is presented in Appendix A.

2.3.2. *TOPSIS*

The TOPSIS method is a type of multi-criteria decision-making technique that was introduced by Hwang and Yoon [157]. The TOPSIS technique is a convenient choice for addressing issues with multiple criteria. In contrast to other methodologies, TOPSIS offers a more practical form of modeling that allows for the inclusion or exclusion of alternative solutions [158].

The TOPSIS method is based on the notion that the optimal alternative should have the greatest geometric distance from the Negative Ideal Solution (NIS) and the shortest distance from the Positive Ideal Solution (PIS) [159]. In the TOPSIS method, it is assumed that the criteria are consistently increasing or decreasing. As the parameters or criteria in this approach are in opposite dimensions, normalization is typically necessary [160,161].

The TOPSIS method involves the following steps:

1. Identify the alternatives and criteria in the decision matrix.
2. Differentiate between qualitative and quantitative criteria.

3. Convert the qualitative criteria into quantitative ones using a bipolar reference scale.
4. Normalize the decision matrix using the norm method, as shown in Equation 3.

$$N = [n_{ij}], n_{ij} = \frac{a_{ij}}{\left[\sum_{i=1}^{m} a_{ij}^2\right]^{\frac{1}{2}}} \tag{4}$$

5. Evaluate the weight of each criterion individually using the Shannon maximum entropy method.
6. Calculate the balanced normalized matrix by multiplying the normalized matrix by a square matrix (Wn*n) where the main diagonal elements correspond to the criterion weights and all other elements are zero.

$$V = N \times W_{n\times n} \tag{5}$$

7. Identify the positive ideal solution (PIS) and negative ideal solution (NIS), and then calculate the geometric distances from each alternative to both solutions. Subsequently, compute the following indices:

Positive ideal solution (V_j^+) = [The vector representing the optimal value of each criterion.]

Negative ideal solution (V_j^-) = [The vector representing the worst value of each criterion.]

Next, perform the following computations:

$$d_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2}, i = 1, 2, ..., m \tag{6}$$

$$d_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2}, i = 1, 2, ..., m \tag{7}$$

The values d_i^+ and d_i^- indicate the geometric distance from the positive ideal and negative ideal, respectively.

8. The relative proximity (RP) can be computed using the following formula:

$$CL_i^* = \frac{d_i^-}{d_i^- + d_i^+} \tag{8}$$

The method that yields the highest RP value is considered to be the most optimal. An illustrative example of the MCDM method TOPSIS is presented in Appendix B.

2.3.3. VIKOR

The VIKOR method is an additional approach to multi-criteria decision-making, which was presented by Opricovic and Tzeng [152]. The VIKOR method is employed for resolving decision problems with conflicting criteria. It consists of the following steps [152,162]:

1. Develop a decision matrix comprising of m alternatives and n criteria (an $m\times n$ matrix).
2. Normalize the matrix using the vector norm. Normalization or scaling is done with the help of vector normalization. The following equation is used for this type of normalization.

$$n_{ij} = \frac{a_{ij}}{\sqrt{\sum_i a_{ij}^2}} \tag{9}$$

3. Compute the weight of each criterion.
4. Calculate the normalized weight matrix.
Calculate the dimensionless weighted matrix (V) using the following formula:

$$V = N \times W_{n\times n} \tag{10}$$

Here, N is a scaleless matrix normalized by vector norming, and W is a matrix of weights for each index.

5. Identify the positive ideal point and negative ideal point.
Determine the maximum (V_j^+) and minimum (V_j^-) values among all alternatives for each criterion. For a positive criterion *i*, we will have:

$$V_j^+ = \max V_{ij} \tag{11}$$

$$V_j^- = \min V_{ij} \tag{12}$$

For negative criteria, the inverse of the above expression is used.

6. Compute the utility (S_i) and regret (R_i) values using the following formula:

$$S_i = L_{1,i} = \sum_{i=1}^{n} \frac{W_j(V_j^* - V_{ij})}{(V_j^* - V_j^-)} \tag{13}$$

$$R_i = L_{\infty,i} = m \left[\sum_{i=1}^{n} W_j(V_j^* - V_{ij})/(V_j^* - V_j^-)\right] \tag{14}$$

Which for positive criteria, $V_j^* = \max V_{ij}, V_j^- = \max V_{ij}$, and W_j is the criterion weight of *j*.

7. Compute the VIKOR index for each alternative (Q_i) using the subsequent formula:

$$Q_i = V \times \frac{(S_i - S^-)}{(S^* - S^-)} + (1 - V) \times \frac{(R_i - R^-)}{(R^* - R^-)} \tag{15}$$

Where $S^- = \min S_i; S^* = \max S_i$ and $R^- = \min R_i$; $R^* = \max R_i$

8. Ranking of Alternatives:
In the VIKOR method, the final ranking of each alternative is determined based on the values of *S*, *R*, and *Q*. The alternative with the lowest values for these three parameters is considered the best.
9. The optimal alternative based on the parameter *Q* must satisfy the following two conditions:

Table 4. The nine-point likert scales.

	1	2	3	4	5	6	7	8	9
Positive Criterias	Strongly Low	Low	Moderately Low	Slightly Low	Moderate	Slightly High	Moderately High	High	Strongly High
Negative Criterias	Strongly High	High	Moderately High	Slightly High	Moderate	Slightly Low	Moderately Low	Low	Strongly Low

Condition 1: Establish the subsequent relationship:

$$Q(A_2) - Q(A_1) \geq \frac{1}{N-1} \tag{16}$$

Where A_1 and A_2 represent the best and second-best alternatives among all options, and N is the number of criteria.

Condition 2: The alternative that ranks first based on the Q parameter must also rank first for at least one of the S or R parameters. If this condition is not satisfied, the ranking will be as follows:

$A_1, A_2, \ldots, A_m$ where A_m is identified according to the subsequent relationship:

$$(A_m) - Q(A_1) < \frac{1}{N-1} \tag{17}$$

If the first condition is not satisfied, then A_1 and A_2 are selected as the optimal alternatives.

2.3.4. WASPAS

Zavadskas, Turskis, Antucheviciene, and Zakarevicius initially presented the WASPAS method [163]. The WASPAS method is a combination of two weighted sum models (WSM) and a weighted product model (WPM). This technique has been demonstrated to exhibit high precision in various studies. The method is adept at tackling both simple and complex optimization problems. Additionally, its straightforward mathematics can be applied in practical scenarios with success and can be effectively employed in decision-making problems [145].

The steps involved in the WASPAS method are as follows:

1. Create a matrix of current status based on the selected indicators.
2. Normalize the status quo matrix. In this method, the status quo matrices can have both positive and negative directions, and Equations (18) and (19) are employed for standardization.

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}} \rightarrow \forall j = 1, 2, \ldots, n \tag{18}$$

$$r_{ij} = \frac{\frac{1}{x_{ij}}}{\sqrt{\sum_{i=1}^{m} \frac{1}{x_{ij}^2}}} \rightarrow \forall j = 1, \ldots, n \tag{19}$$

3. Compute the weight of each indicator.
4. Estimate the variance of the normalized values using the subsequent formula:

$$\sigma^2(\bar{x_{ij}}) = (0.05\, x_{ij})^2 \tag{20}$$

5. Calculate the variance of $\sigma^2\left(\sigma_i^{(1)}\right)$ and $\sigma^2\left(\sigma_i^{(2)}\right)$ using the following equations:

$$\sigma^2\left(\sigma_i^{(1)}\right) = \sum_{j=1}^{n} \bar{x_{ij}}\, w_j^2 \cdot \sigma^2(\bar{x_{ij}}) \tag{21}$$

$$\sigma^2\left(\sigma_i^{(2)}\right) = \sum_{j=1}^{n} \left[\frac{\prod_{j=1}^{n} (\bar{x_{ij}})^{wj} \times w_{ij}}{(x_{ij})^{wj} (\bar{x_{ij}})^{1-wj}}\right] \cdot \sigma^2(x_{ij}) \tag{22}$$

6. Compute the values of λ and Q_i to rank the alternatives using the following equations:

$$\lambda = \frac{\sigma^2\left(\sigma_i^{(2)}\right)}{\sigma^2\left(\sigma_i^{(1)}\right) + \sigma^2\left(\sigma_i^{(2)}\right)} \tag{23}$$

$$Q_i = \lambda \sum_{j=1}^{n} \bar{x_{ij}}\, w_j + (1-\lambda) \prod_{j=1}^{n} (x_{ij})^{wj}, \lambda = 0, \ldots, 1 \tag{24}$$

The alternative with a higher Q_i value is considered to be the better option.

3. Results and discussion

Table 5 provides the data for the various alternatives and criteria related to lunar habitat construction, which were gathered through a review of existing literature. The criteria are color-coded: negative factors are marked in red, while positive factors are highlighted in green. Positive criteria represent elements that improve the system's performance, whereas negative criteria reflect factors that reduce its efficiency. All the references mentioned in this research have been used to extract the required parameters. In fact, all the results mentioned in the research have been examined and the desired values and quantities have been placed in the Table 4 after consulting experienced experts. To quantify the qualitative data, we used the Likert scale, a bipolar reference scale, to convert the values into numerical terms [164].

3.1. EWM to determine the weights of criteria

As shown in Figure 3, the EWM analysis revealed that density had the lowest weight, while temperature had the highest weight. These weights reflect the relative importance of each criterion in the decision-making process.

3.2. TOPSIS for lunar concrete selection

The results of the TOPSIS analysis, based on the relative proximity (CL) measure, are shown in Table 6. According to the table, the best concrete mortar for lunar habitat

Table 5. Decision matrix for selecting the best concrete for lunar habitat construction [1–8,14–22].

	Ordinary Portland cement	Sulfur Concrete	Geopolymer Concrete	Sintered materials	Polymer-bound regolith Concrete	Products of geo-thermite reactions	Regolith-based magnesium oxychloride	Microbial induced calcite precipitation
Shipping	Strongly High	Strongly High	Strongly Low	Moderately High	Moderately Low	Slightly High	Moderately Low	Slightly High
Water requirement	Moderately High	Strongly Low	Moderate	Strongly Low	Low	Low	Low	Moderate
Technical working conditions	Moderately Low	Moderate	Moderate	Moderate	Moderate	Moderate	Strongly High	Strongly High
Curing time	Moderately High	Strongly Low	Moderately Low	Strongly Low	Moderate	Moderately Low	Moderately High	Moderate
Temperature (°C)	1450	900–1200	20–90	900–1400	−100 To 400	900–1900	20–800	−20 To +50
Total required energy	Moderately High	Moderately High	Moderate	Strongly High	Moderate	Moderately High	High	Moderately Low
Availability	Slightly Low	Strongly Low	Moderately High	Strongly High	Moderately Low	Moderate	Moderate	Strongly Low
Strength (MPa)	15–35	20–65	16–36	2–60	20–100	10–18	20–80	1–5
Durability	Moderately High	Moderate	High	Strongly High	Moderate	Moderately Low	Slightly High	Low
Cosmic radiation shield: Density (gr/cm^3)	2.4	2.24	2.46	2.1	1.4	1.7	2.5	1.56
Cosmic radiation shield: Hydrogen content	Moderately Low	Strongly Low	Slightly Low	High	Slightly High	Moderately Low	Moderate	Low
Additives needed	High	Low	Moderately High	Strongly High	Slightly High	Moderate	Moderate	Strongly Low
Sustainability	Moderate	Slightly High	High	Strongly High	Moderately Low	Slightly Low	Slightly High	Moderately Low
Safety	Slightly High	Moderate	Moderately High	Strongly High	Slightly High	Moderate	Moderately High	Slightly Low
Recyclability	Moderately Low	Moderately High	Slightly High	High	Slightly Low	Slightly Low	Moderately Low	Low

Table 6. Ranking concretes based on the TOPSIS method.

Concretes	TOPSIS Ranking
Ordinary Portland cement	7
Sulfur Concrete	6
Geopolymer Concrete	2
Sintered materials	4
Polymer-bound regolith Concrete	1
Products of geo-thermite reactions	8
Regolith-based magnesium oxychloride	3
Microbial induced calcite precipitation	5

construction is Polymer-bound regolith, while the least favorable option is the Products of geo-thermite reactions.

3.3. *VIKOR for lunar concrete selection*

According to the final stage of the VIKOR method, alternatives are ranked in ascending order based on their values in the S, R, and Q matrices. Ideally, the best compromise solution is the one that ranks first in all three lists—meaning it has the lowest S, R, and Q values simultaneously. However, such a dominant alternative does not exist in our results.

In the absence of a clearly dominant option, the VIKOR methodology prescribes an evaluation of two specific conditions to identify the compromise solution(s) based on Q-values. In our case, Regolith-based magnesium oxychloride and Polymer-bound regolith are ranked first and second in terms of Q, respectively. Upon further examination, Condition 1 (acceptable advantage) is not satisfied, as the difference between the Q-values of the top two alternatives is smaller than the acceptable threshold. Condition 2 (acceptable stability in decision making), however, is met.

As Condition 1 fails, the VIKOR method recommends selecting a set of alternatives as compromise solutions, rather than a single best one. Following this approach, the largest subset of top-performing alternatives with identical lowest Q-values is identified. In our study, this results in three alternatives—Sintered materials, Polymer-bound regolith, and Regolith-based magnesium oxychloride—being jointly selected as compromise solutions. The remaining options are then ranked accordingly based on their Q-values.

On the other hand, the least favorable concrete mortar for lunar habitat construction is Microbially induced calcite precipitation (Table 7).

3.4. *WASPAS for lunar concrete selection*

According to the WASPAS methodology, a higher Q_i value indicates a better alternative. As demonstrated in Table 8, the optimal and least favorable concrete mortars for lunar habitat construction are Sintered materials and Products of geo-thermite reactions, respectively.

3.5. *Average*

The three MCDM techniques—TOPSIS, VIKOR, and WASPAS—produced slightly different results due to

Table 7. Ranking concretes based on the VIKOR method.

Concretes	VIKOR Ranking
Ordinary Portland cement	3
Sulfur Concrete	2
Geopolymer Concrete	2
Sintered materials	1
Polymer-bound regolith Concrete	1
Products of geo-thermite reactions	5
Regolith-based magnesium oxychloride	1
Microbial induced calcite precipitation	4

Table 8. Ranking concretes based on the WASPAS method.

Concretes	WASPAS Ranking
Ordinary Portland cement	4
Sulfur Concrete	7
Geopolymer Concrete	2
Sintered materials	1
Polymer-bound regolith Concrete	5
Products of geo-thermite reactions	8
Regolith-based magnesium oxychloride	3
Microbial induced calcite precipitation	6

variations in scoring scales, weightings, and score distributions [165]. To determine the best concrete mortar for the Moon, this study used the average, which is based on the average scores across all approaches [166]. Table 9 shows the results of the three MCDM methods and the corresponding average scores for the various concrete options. According to this analysis, geopolymer and sintered materials are considered the most suitable concretes for the Moon's challenging conditions. In contrast, products of geo-thermite reactions are identified as the least favorable option for lunar habitat construction.

Although numerous studies have explored space concrete, most have focused either on comparing different materials or on a limited set of key parameters for concrete construction [117]. While each type of concrete may offer unique advantages for specific applications, the optimal choice must be versatile enough to perform well in a variety of situations. Additionally, given the high costs and investment risks associated with space research, all relevant factors must be considered during the decision-making process [164]. Despite the use of various analytical methods, laboratory tests under diverse simulated conditions remain the most reliable approach for identifying the best concrete for space habitat construction. When feasible and economically viable, empirical analysis should be prioritized, as it provides more accurate and reliable data, enabling better-informed decisions about the selection of materials for space construction. Therefore, whenever possible and cost-effective, empirical testing should be the preferred method for determining the best concrete for space habitat construction [167]. The findings of this study may evolve as further research is conducted on the concrete materials examined. The results could be adjusted by revising the weightings and subjective probabilities used in the MCDM analysis.

Table 9. Ranking of concretes based on the three different MCDM methods and the average score.

Concretes	VIKOR Ranking	WASPAS Ranking	TOPSIS Ranking	Final Rank
Ordinary Portland cement	7	3	4	3
Sulfur Concrete	6	2	7	4
Geopolymer Concrete	2	2	2	1
Sintered materials	4	1	1	1
Polymer-bound regolith Concrete	1	1	5	2
Products of geo-thermite reactions	8	5	8	5
Regolith-based magnesium oxychloride	3	1	3	2
Microbial induced calcite precipitation	5	4	6	4

Additionally, incorporating uncertainty, probability, and neural networks into MCDM studies could enhance the accuracy of decision-making processes, making these techniques valuable for future research. Using these advanced methods can help mitigate risks associated with space construction and improve the overall success of space exploration missions [168]. Furthermore, the decision matrix could be expanded to include other important factors, such as costs. However, further economic studies are necessary to assess the cost-effectiveness of different concrete options. Including economic considerations in the decision-making process is essential to ensure the financial sustainability and long-term feasibility of space construction projects.

3.6. Regolith properties effect on final selection

This investigation primarily examines the Moon's extreme environmental conditions, including its ultrahigh vacuum and cryogenic temperatures, while acknowledging the understudied aspect of regolith compositional variability. To address this research gap, we evaluate three candidate construction materials—geopolymer concrete, sulfur concrete, and sintered regolith—with particular attention to their materials compatibility with lunar surface composition. Geopolymer concrete formulations require aluminosilicate-rich glassy phases, with recent advances in material modification demonstrating potential pathways for optimizing lunar regolith suitability [169]. Comparative studies utilizing lunar regolith simulants have confirmed the exceptional mechanical performance of these geopolymer binders [170]. The presence of calcium oxide phases in the regolith may provide additional strength enhancement through the formation of calcium-aluminosilicate-hydrate networks [171], though the economic viability of processing calcium-rich feedstocks requires careful consideration [172]. Sulfur-based concrete systems present distinct challenges in the lunar environment. While extant sulfur deposits (~1 wt% surface concentration) could theoretically support this technology, the material's thermodynamic instability under ultrahigh vacuum conditions raises significant implementation concerns [121]. These challenges are compounded by the complex dynamics of the lunar exosphere and its interaction with surface materials [173]. Sintered regolith architectures offer a potentially more robust solution, eliminating binder requirements through direct particulate consolidation. Recent multicriteria decision analyses of extraterrestrial construction materials have identified sintering as a leading candidate technology, particularly when evaluating implementation feasibility across multiple planetary surfaces [24]. Recent comparative analyses of in-situ construction technologies have provided critical insights into optimizing material selection for lunar habitats [13]. These studies emphasize the need for materials that not only withstand extreme environmental conditions but also accommodate the logistical constraints of robotic construction systems. Particularly promising are developments in rheology-modifying additives, where innovations in time-release polycarboxylate ether (PCE) design have demonstrated significant potential for controlling workability in cementitious systems [174]. Such advances could prove transformative for lunar geopolymer applications, potentially enabling extended working times and improved placement characteristics under vacuum conditions while maintaining the mechanical performance requirements for habitat structures. In conclusion, both geopolymer and sintered regolith systems demonstrate significant potential for lunar construction applications, though each presents distinct materials science challenges that must be resolved through continued investigation. Optimal material selection will require comprehensive performance evaluation under simulated lunar conditions, coupled with rigorous systems-level analysis of implementation pathways.

4. Conclusion

- The goal of this study was to determine the most suitable concrete mortar for building habitats on the Moon, considering both technical and economic factors. To accomplish this, three different multiple criteria decision-making (MCDM) methods like TOPSIS, VIKOR, and WASPAS were utilized.
- The results from these methods were quite similar, and by averaging the outcomes, the final ranking was established. According to this approach, geopolymer and sintered materials emerged as the top construction materials, based on the average ratings, without factoring in the variations in regolith composition.
- Polymer-bound regolith and regolith-based magnesium oxychloride were ranked second, while Portland cement, sulfur, microbial-induced calcite precipitation, and products of geo-thermite reactions were placed lower in the ranking.

- The study also explored how regolith composition affects material selection, considering factors like mineralogy, morphology, sulfur content, and calcium oxide (CaO). It was concluded that geopolymer and sintered materials are both suitable options for construction on the Moon.
- Additionally, the Entropy method was employed to identify that temperature and density were the most and least influential criteria, respectively, in the decision-making process.
- In conclusion, this study provided a systematic approach for comparing various types of concrete, which can be refined and expanded upon to improve the accuracy of decision-making in selecting space construction materials.

Acknowledge AI usage

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No potential conflict of interest was reported by the author(s).

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Appendix A

An example for the Entropy Weight Method is provided below.

	C1	C2	C3
A1	90	22	Strongly High
A2	85	24	Strongly High
A3	82	23	Moderately High
A4	75	25	Moderate

	C1	C2	C3
A1	90	22	9
A2	85	24	9
A3	82	23	7
A4	75	25	5

$$P_{ij} = \frac{x_{ij}}{\sum_{i=1}^{m} x_{ij}}$$

	C1	C2	C3
A1	0.271	0.234	0.300
A2	0.256	0.255	0.300
A3	0.247	0.245	0.233
A4	0.226	0.266	0.167

$$E_j = -\frac{1}{Ln(m)} \sum_{i=1}^{m} P_{ij} Ln(P_{ij})$$

	C1	C2	C3
E_j	0.9984	0.9992	0.9814

$$d_j = 1 - E_j$$

	C1	C2	C3
Wj	0.0743	0.0390	0.8866

$$w_j = \frac{d_j}{\sum_{j=1}^{n} d_j}$$

	C1	C2	C3
d_j	0.0016	0.0008	0.0186

Appendix B

An example for one MCDM methods (TOPSIS) is provided below.

Steps

- Construct the decision matrix: List all alternatives and criteria.

Alternatives: A1, A2, A3, A4 Criteria: C1 (−), C2 (−), C3 (+)

The positive and negative signs for each metric indicate whether its increase is profitable (+) or detrimental (−).

	C1	C2	C2
A1	300	20	25
A2	400	25	30
A3	200	30	40
A4	500	35	45
weight	0.45	0.3	0.25

- Normalize the decision matrix: Use vector normalization to transform criteria into a comparable scale.

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}$$

- Determine the weighted normalized decision matrix: Multiply normalized values by their respective weights.

- Identify the positive ideal solution (PIS) and negative ideal solution (NIS: PIS is the best value for each criterion, and NIS is the worst.

- Calculate the separation measures: Compute the distance of each alternative from PIS and NIS.

$$s_i^* = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^*\right)^2}$$

$$s_i^- = \sqrt{\sum_{j=1}^{n} \left(v_{ij} - v_j^-\right)^2}$$

- Determine the relative closeness to the ideal solution: Calculate the closeness coefficient for each alternative.
- Rank the alternatives: Rank based on the closeness coefficient, with higher values indicating better alternatives.

	C1	C2	C2
A1	0.408	0.356	0.348
A2	0.544	0.445	0.418
A3	0.272	0.535	0.557
A4	0.680	0.624	0.627

	C1	C2	C2
A1	0.184	0.107	0.087
A2	0.245	0.134	0.105
A3	0.122	0.160	0.139
A4	0.306	0.187	0.157

	C1	C2	C2
A*	0.122	0.107	0.157
A-	0.306	0.187	0.087

$$C_i^* = \frac{s_i^-}{s_i^* + s_i^-}$$

	C1	C2	C2
A1	0.004	0.000	0.005
A2	0.015	0.001	0.003
A3	0.000	0.003	0.000
A4	0.034	0.006	0.000

Example: Consider selecting a generator based on cri-

	C1	C2	C2
A1	0.015	0.006	0.000
A2	0.004	0.003	0.000
A3	0.034	0.001	0.003
A4	0.000	0.000	0.005

	Si*	Si⁻
A1	0.093	0.146
A2	0.136	0.083
A3	0.056	0.193
A4	0.200	0.070

	Ci*	Ranking
A1	0.612	2
A2	0.380	3
A3	0.774	1
A4	0.258	4

teria such as power consumption, fuel tank capacity, and fuel consumption. After constructing and normalizing the decision matrix, the distance to the PIS and NIS are calculated, and then accordingly ranked among those generators [175–179].